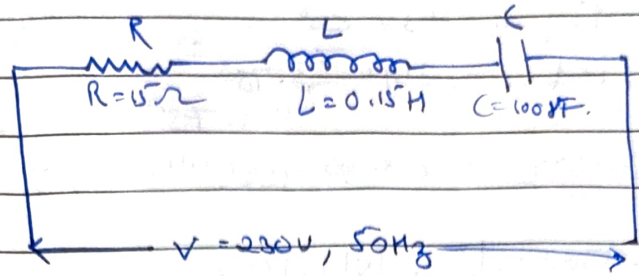
Name:- Rituik Mahajan
25BCE10509

Quick Work
Page No.:
Date: 30/09/2025
M T W T F S S

EEE

Assignment - 2

Q1. Sol:-



Given:-

$R = 15\,\Omega$

$L = 0.15\,H$

$C = 100 \times 10^{-6}\,F.$

⇒ For current,

$Z = R + jX$ where $X = X_L - X_C$.

$X_L = \omega L.$

$X_L = 2\pi(50)(0.15)$

$X_L = 100\pi(0.15)$

$\boxed{X_L = 47.12\,\Omega}$

$X_C = \frac{1}{\omega C}$

$X_C = \frac{1}{2\pi f C} \Rightarrow \frac{1}{0.31} \Rightarrow \boxed{3.225\,\Omega}$

⇒ $|Z| = \sqrt{(R)^2 + (X_L - X_C)^2}\ \tan^{-1}\left(\frac{X_L - X_C}{R}\right)$

$|Z| = \sqrt{225 + 43.9}\ \tan^{-1}\left(\frac{43.9}{225}\right)$

$|Z| = \sqrt{268.9}\ \angle 11.04^\circ$

$\boxed{|Z| = 16.39\,\Omega \angle 11.04^\circ}$

⇒ $|Z| = \sqrt{225 + (14.87)^2}\ \tan^{-1}\left(\frac{14.87}{225}\right)$

$|Z| = \sqrt{225 + 221.11}\ \angle 3.78^\circ$

$\boxed{|Z| = 21.12\,\Omega \angle 3.78^\circ}$

Quick Work

Page No.:

Date:

M T W T F S S

=) So current is :-

$I = \frac{E}{|Z|} \Rightarrow \frac{230\angle 0^\circ}{21.12\angle 3.75^\circ} \Rightarrow \boxed{10.84A \angle -34.75^\circ}$ Ans

44.75

⇒ Power factor = cos θ ⇒ cos(44.75°) ⇒ $\boxed{0.71 \text{ watts}}$ lagging

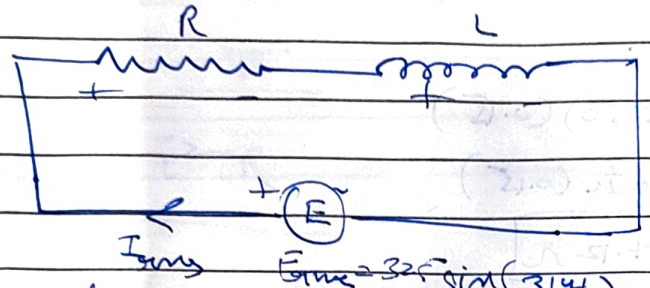
Power consumed = |V||I| cos θ

⇒ (230)(10.84) cos 44.75°

⇒ $\boxed{1770.17 \text{ watts}}$ Ans

Q.2. Sol:. $V_{rms} = 325 \sin(314t)$

$I_{rms} = 14.14 \sin(314t - 60^\circ)$



⇒ the eqn for current is :-

$I_{rms} = I_0 \sin(\omega t)$

$V_{rms} = V_0 \sin(\omega t - \phi)$

So here,

$I_0 = 14.14 A$

$V_0 = 325 V.$

$\omega = 314$ rad/s

$\phi = 60^\circ.$

$V_{rms} = \frac{V_0}{\sqrt{2}} \Rightarrow \frac{325}{\sqrt{2}} = 230 V$

$I_{rms} = \frac{I_0}{\sqrt{2}} = \frac{14.14}{\sqrt{2}} = 10 A.$

⇒ So we know,

$V_{rms} = I_{rms} R$

$R = \frac{V_0}{I_0}$

$R = \frac{325}{14.14} \Rightarrow \boxed{22.98 \Omega}$

$R = \frac{230}{10} \Rightarrow \boxed{23 \Omega}$

→ for resistance

$\cos\phi = \frac{R}{Z}$ $\phi = \tan^{-1}\left(\frac{X_L}{R}\right) \Rightarrow \tan^{-1}(60^\circ)$

$R = Z\cos\phi$

$R = Z\cos 60$

$R = (23)(0.5) \Rightarrow \boxed{11.5\,\Omega}$

for inductor,

→ $X_L = \omega L$,

$X_L = Z\sin\theta$.

$\Rightarrow X_L = 23(0.866) \Rightarrow \boxed{19.9\,\Omega}$

$\Rightarrow \frac{19.9}{314} = L$.

$\boxed{L = 0.06344\,H}$

Power consumed $= V_{rms} I_{rms} \cos\phi$

$\Rightarrow (230)(10)\cos 60^\circ$

$\Rightarrow \frac{2300 \times 1}{2}$

$\boxed{1150 \text{ watts}}$

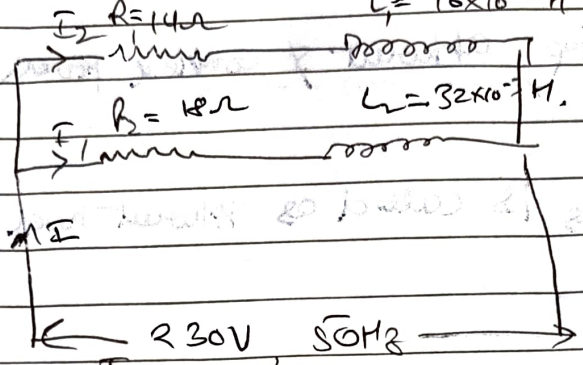
Quick Work

Page No.:

Date:

M T W T F S S

Q.3. Sol:-



for current I_1,

$\Rightarrow X_L = \omega L$

$X_L = 2\pi(50)(32\times10^{-3})$

$X_L = 100\pi\,(32\times10^{-3})$

$X_L = \frac{32\times\pi}{10}$

$X_L = 3.2\pi$

$\boxed{X_L = 10.05\,\Omega}$

$\Rightarrow Z_2 = R + jX$ where $X = X_L$

$\Rightarrow |Z_2| = \sqrt{R^2 + X_L^2}\ \angle \tan^{-1}\left(\frac{X_L}{R}\right)$

$|Z_2| = \sqrt{(18)^2 + (10.05)^2}\ \angle \tan^{-1}\left(\frac{10.05}{18}\right)$

$|Z_2| = \sqrt{324 + 101.0025}\ \angle 29.17^\circ$

$|Z_2| = \sqrt{425.0025}\ \angle 29.17^\circ$

$\boxed{|Z_2| = 20.61\,\Omega \angle 29.17^\circ}$

$\Rightarrow$ so current I_1,

$\Rightarrow I_1 = \frac{E}{|Z_2|}$

$I_1 = \frac{230\angle 0^\circ}{20.61\angle 29.17^\circ} \Rightarrow \boxed{11.15\,A \angle -29.17^\circ}$

Quick Work

Page No.:

Date:

M T W T F S S

EEE Assignment continued

=> For current I_2,

=> $X_L = \omega L$.

$X_L = 2\pi(50)(16\times10^{-3})$

$X_L = 100\pi\ 16\times\frac{1}{1000}$.

$X_L = 1.6\pi$

$\boxed{X_L = 5.02\Omega}$

=> $|Z_1| = \sqrt{(14)^2+(5.02)^2}\ \tan^{-1}\left(\frac{5.02}{14}\right)$

$|Z_1| = \sqrt{196+25.2004}\ \angle 19.72^\circ$

$|Z_1| = \sqrt{221.2004}\ \angle 19.72^\circ$

$\boxed{|Z_1| = 14.87\Omega\ \angle 19.72^\circ}$

So current I_2,

$I_2 = \frac{E}{|Z_1|} \Rightarrow \frac{230\angle 0^\circ}{14.87\angle 19.72^\circ} \Rightarrow \boxed{15.46\Omega\angle -19.72^\circ}$

=) For line current

$I_{total} = I_1 + I_2 = 15.46 + 11.15 \angle -23.$

$\Rightarrow \boxed{26.61 \angle -23^\circ}$

Total power factor - $\cos(\phi_{total})$

$\Rightarrow \cos(-23^\circ) \Rightarrow$

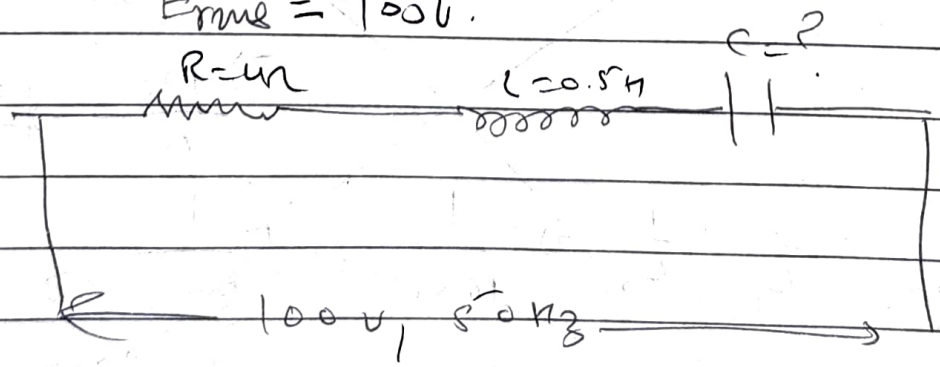
$\Rightarrow \boxed{0.92 \text{ lagging}}$

Q.4. Sol:- $R = 4\Omega$

$L = 0.5H$

$f = 50Hz.$

$E_{rms} = 100V.$



(a)- Sol:- For resonance,

$X_L = X_C$

so impedance

$Z = R.$

Quick Work

Page No.:

Date:

M T W T F S S

$\Rightarrow X_L = \omega L$

$\Rightarrow X_L = 2\pi f L$

$X_L = 2\pi(50)(0.5)$

$X_L = 100\pi\left(\frac{0.5}{10}\right)$

$X_L = 50\pi$

$\boxed{X_L = 157.07\,\Omega}$

since $X_L = X_C$

$X_C = \frac{1}{\omega C}$

$157.07 = \frac{1}{2\pi(50)C}$

$C = \frac{1}{(157.07)(100\pi)}$

$C = \frac{1}{49344.99}$

$\boxed{C = 2.02 \times 10^{-5}\,F}$

(b) sol:- $V_C =$

For current at resonance

$\Rightarrow I = \frac{E_{rms}}{R} = \frac{100\sqrt{2}}{4\sqrt{2}} = 25\,A$

$\Rightarrow$ so V_C is

$V_C = I(X_C)$

$V_C = 25(157.07)$

$\boxed{V_C = 3926.75\,V}$

Quick Work

Page No.:

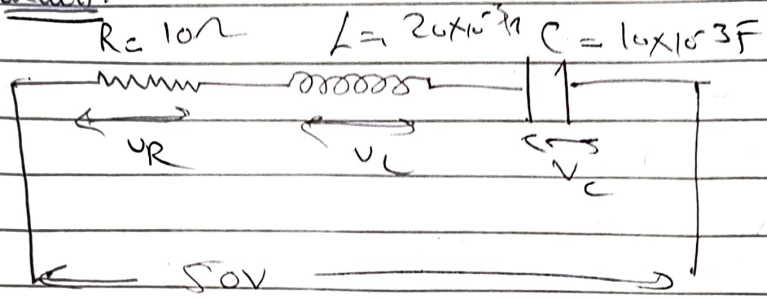
Date:

M T W T F S S

$\Rightarrow$ Q-factor $= \frac{V_c}{V} \Rightarrow \frac{3926.75}{100} \Rightarrow \boxed{39.26}$

Q.5 Sol:-

Circuit:-



$\Rightarrow$ For resonant frequency,

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$f_0 = \frac{1}{2\pi\sqrt{(20\times10^{-3})(10\times10^{-3})}}$$

$$f_0 = \frac{1}{2\pi\sqrt{200\times10^{-6}}}$$

$$f_0 = \frac{1}{2\pi\sqrt{\frac{200}{1000000}}}$$

$$f_0 = \frac{\sqrt{5000}}{2\pi}$$

$$f_0 = \frac{70.71}{6.28}$$

$$\boxed{f_0 = 11.25\ Hz}$$

$\rightarrow$ for V_R,

$V_R = I(R)$.

$\Rightarrow$ At resonant frequency

$X_L = X_C$.

$Z = R$.

Quick Work

Page No.:

Date:

M T W T F S S

⇒ & current would be,

$V_{rms} = I(Z)$

$I = \frac{50}{10}$

$\boxed{I = 5A}$

so V_R

$V_R = 5(10)$

$\boxed{V_R = 50V}$

for V_L

$V_L = I(X_L)$

& $X_L = \omega L$.

$\omega = 2\pi f_0$

$X_L = 2\pi f_0 L$.

$X_L = 2\pi(11.25)(20\times10^{-3})$

$X_L = 1413.71 \times 10^{-3}$.

$X_L = \frac{1413.71}{1000}$.

$\boxed{X_L = 1.41\Omega}$

⇒ so $V_L = 5(1.41)$

$\boxed{V_L = 7.05V}$

Similarly

⇒ $\boxed{V_C = 7.05V}$ as $(X_C = X_L)$.

Q factor $= \frac{V_C}{V} \Rightarrow \frac{7.05}{50} \Rightarrow \boxed{0.141}$ Ans.

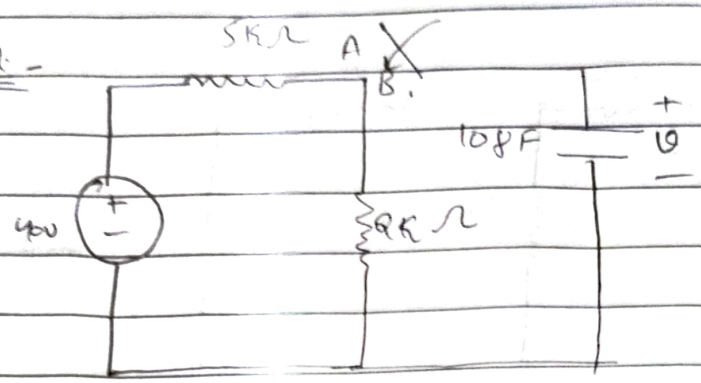
Quick Work

Page No.:

Date:

M T W T F S S

Q6 Sol:-



⇒ After switch closed,

$R_{eq} = 2k\Omega$

$\tau = R_{eq} C$

$\tau = (2\times10^3)(10\times10^{-6})$

$\tau = 20\times10^{-3}$

$\boxed{\tau = 2\times10^{-2}\,s}$

⇒ $v(t) = V_0 e^{-t/\tau}$ where $\tau = R_{eq}C$

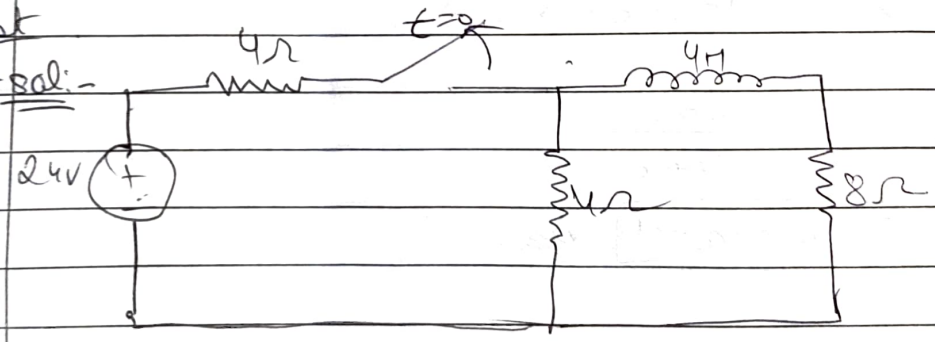
$v(t) = 40e^{-t/2\times10^{-2}}$

$v(t) = 40e^{-t\times100/2}$

$\boxed{v(t) = 40e^{-50t}}$ Ans

Recent

Q-7 Sol:-



⇒ $R_{eq} = \frac{1}{4} + \frac{1}{8} \Rightarrow \frac{2+1}{8} \Rightarrow \frac{3}{8} = \frac{8}{3}\Omega$.

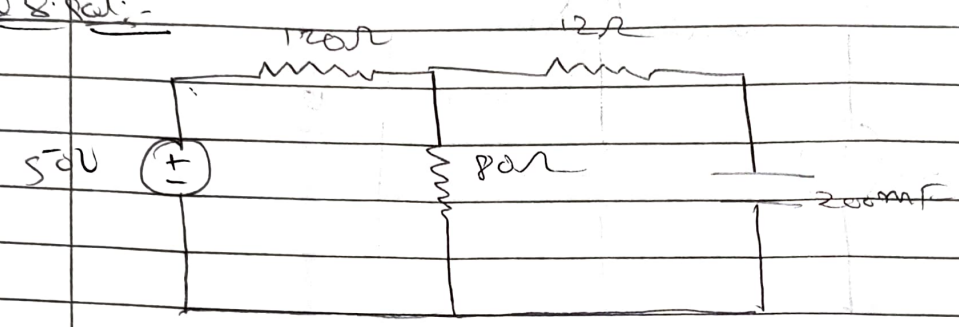
Quick Work

Page No.:

Date:

M T W T F S S

Q 8: Sol:-



⇒ for time constant,

$\tau = R_{eq} C$.

$\Rightarrow \frac{1}{R_p} = \frac{1 \times 2}{120} + \frac{1 \times 3}{80} \Rightarrow \frac{2+3}{240}$.

$R_p = \frac{240}{5}$.

$R_p = 48$

$R_{eq} = R_p + R_{12} \Rightarrow 48 + 12 \Rightarrow \boxed{60\Omega}$

$\Rightarrow \tau = 60 (200 \times 10^{-3})$

$\tau = 12000 \times 10^{-3}$

$\tau = \frac{12000}{1000}$

$\boxed{\tau = 12 \text{ seconds}}$ Ans

2 | 48
2 | 60, 40
2 | 30, 20
3 | 15, 10
5 | 5, 5
3 | 1, 1
48
24
12
6